

FIFA 21 PLAYER MARKET & PERFORMANCE ANALYSIS

Data Cleaning & Prep Report

- PRE-STEP 1: Fix & Expand Column Names

Renamed abbreviated or inconsistent column names to clear, descriptive labels to improve readability, understanding, and ease of analysis.

CODE:

```
In [2]: # --- PRE-STEP 1: Fix & Expand Column Names
rename_dict = {
    'POT': 'Potential',
    'OVA': 'Overall',
    '↓OVA': 'Overall',
    'BOV': 'Best_Overall',
    'A/W': 'Attack_Work_Rate',
    'D/W': 'Defensive_Work_Rate',
    'W/F': 'Weak_Foot',
    'SM': 'Skill_Moves',
    'IR': 'International_Reputation'
}
df = df.rename(columns=rename_dict)
```

- PRE-STEP 2: Drop Useless Columns (Case-Sensitive!)

Removed unnecessary or irrelevant columns that do not contribute to the analysis, reducing dataset complexity and improving processing efficiency.

CODE:

```
columns_to_drop = ['playerUrl', 'photoUrl', 'Loan Date End']
df = df.drop(columns=columns_to_drop, errors='ignore')
```

- PART A: Clean Financial Data

Cleaned and standardized financial data by handling missing values, correcting formats, and ensuring consistency for accurate analysis and reporting.

CODE:

```
def clean_currency(x):
    if isinstance(x, str):
        x = x.replace('€', '').strip()
        if 'M' in x:
            x = float(x.replace('M', '')) * 1000000
        elif 'K' in x:
            x = float(x.replace('K', '')) * 1000
        else:
            x = float(x)
    return float(x)

financial_cols = ['Value', 'Wage', 'Release Clause']
for col in financial_cols:
    df[col] = df[col].fillna('0')
    df[col] = df[col].apply(clean_currency)

# Create Moneyball Features
df['Wage_to_Value_Ratio'] = df['Wage'] / (df['Value'] + 1)
df['Potential_Growth'] = df['Potential'] - df['Overall']
```

- PART B: Clean Hits and Star Ratings

Converted Hits and Star Rating values into consistent numeric formats by removing symbols, handling text values, and preparing them for accurate analysis and visualization.

CODE:

```
def clean_hits(x):
    if pd.isna(x):
        return 0
    if isinstance(x, str):
        x = x.replace(',', '')
        if 'K' in x:
            return float(x.replace('K', '')) * 1000
        else:
            return float(x)
    return float(x)

df['Hits'] = df['Hits'].apply(clean_hits)

star_columns = ['Weak_Foot', 'Skill_Moves', 'International_Reputation']
for col in star_columns:
    df[col] = df[col].astype(str).str.replace('★', '', regex=False).str.strip().astype(int)
```

- PART C: Clean Physical Traits & Contracts

Standardized physical trait and contract columns by extracting numeric values, correcting formats, and ensuring consistent data for reliable analysis.

CODE:

```
def clean_height(x):
    if isinstance(x, str):
        if 'cm' in x:
            return int(x.replace('cm', ''))
        elif "'" in x:
            parts = x.split("'")
            feet = int(parts[0])
            inches = int(parts[1].replace("'", ''))
            return round((feet * 12 + inches) * 2.54)
    return x

df['Height'] = df['Height'].apply(clean_height)

def clean_weight(x):
    if isinstance(x, str):
        if 'kg' in x:
            return int(x.replace('kg', ''))
        elif 'lbs' in x:
            lbs = int(x.replace('lbs', ''))
            return round(lbs / 2.20462)
    return x

df['Weight'] = df['Weight'].apply(clean_weight)

def extract_contract_end(x):
    if isinstance(x, str):
        if '~' in x:
            return x.split('~')[1].strip()
        elif 'On Loan' in x:
            return x[-12:-8]
        elif 'Free' in x:
            return 'Free'
    return str(x)

df['Contract_Valid_Until'] = df['Contract'].apply(extract_contract_end)
```

- FINAL HOUSEKEEPING: Drop Redundant Columns

Removed redundant and intermediate columns created during data cleaning to produce a streamlined, analysis-ready dataset.

CODE:

```
df = df.drop(columns=['LongName', 'Contract'], errors='ignore')

# --- FINAL CHECK ---
print("--- Master Cleaning Complete! ---")
print(f"Final Dataset Shape: {df.shape}")
print("\nSample of Newly Cleaned Columns:")
print(df[['Name', 'Skill_Moves', 'Height', 'Weight', 'Contract_Valid_Until']].head())
```

OUTPUT:

```
--- Master Cleaning Complete! ---
Final Dataset Shape: (18979, 75)
```

Sample of Newly Cleaned Columns:

	Name	Skill_Moves	Height	Weight	Contract_Valid_Until
0	L. Messi	4	170	72	2021
1	Cristiano Ronaldo	5	187	83	2022
2	J. Oblak	1	188	87	2023
3	K. De Bruyne	4	181	70	2023
4	Neymar Jr	5	175	68	2022